

1. a. Correct.

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 3 & 5 & 6 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

so $2(1, 2, 3) = (2, 4, 6)$, meaning that $\mathbf{v}_2$ can be written as a linear combination of $\mathbf{v}_1$. Thus $2\mathbf{v}_1 - \mathbf{v}_3 = \mathbf{0}$ is a nontrivial linear combination that equals $\mathbf{0}$.

- b. Incorrect. $\mathbf{v}_2$ cannot be written as a combination of $\mathbf{v}_1$ and $\mathbf{v}_3$. To see this solve the augmented matrix,

$$\left[\begin{array}{cc|c} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 3 & 6 & 5 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

which is inconsistent.

2. a. Yes. Suppose that $c_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 9 \end{bmatrix} = \mathbf{0}$

$$\left[\begin{array}{cc} 1 & 2 \\ 4 & 9 \end{array} \right] \rightsquigarrow \text{R2-4R1} \left[\begin{array}{cc} 1 & 2 \\ 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right]$$

implies that $c_1 = c_2 = 0$. Thus the vectors are linearly independent.

- d. As above we need to put the associated column matrix into reduced echelon form.

$$\left[\begin{array}{ccc} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 1 & 3 & 2 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc} 1 & 2 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

so the vectors are linearly independent. Note that you only need to reduce the matrix to

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

where it is clear the matrix has three pivots and thus a rank of 3.

3. The span of the vectors need to be equal to the given set, and the set needs to be linearly independent.

- b. No.

$$\left[\begin{array}{cccc} 1 & 1 & 2 & 2 \\ 0 & 2 & 2 & 2 \\ 1 & 4 & 5 & -1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

So there is a non-trivial c_1, c_2, c_3, c_4 such that $c_1(1, 0, 1) + c_2(1, 2, 4) + c_3(2, 2, 5) + c_4(2, 2, -1) = \mathbf{0}$. To find it, $c_1 = -c_3$, $c_2 = -c_3$, and $c_4 = 0$. For $c_3 = -1$ then $1(1, 0, 1) + 1(1, 2, 4) - (2, 2, 5) + 0(2, 2, -1) = \mathbf{0}$.

Alternatively, by proposition 3.4 if the column vectors from $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 0 & 2 & 2 & 2 \\ 1 & 4 & 5 & -1 \end{bmatrix}$ are a basis of $\mathbb{R}^3$

then it is a 3×3 nonsingular matrix but it is a 3×4 .

Any four vectors in $\mathbb{R}^3$ form a linearly dependent set. In general, any n vectors in $\mathbb{R}^{n-1}$ (for $n > 1$) form a linearly dependent set. This is a result of proposition 4.1 or 3.4, but you can see this by evaluating the possible reduced echelon forms of the augmented column matrix,

$$A\mathbf{x} = \mathbf{b} = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & a_{1,n+1} \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & a_{n,n+1} \end{array} \right].$$

An inconsistent matrix means the vectors composing A are linearly dependent, but a solvable matrix means that $\mathbf{b}$ can be written as a linear combination of the vectors composing A . Either way the set of vectors is linearly dependent.

c. No.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 4 \\ 3 & 1 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Although these vectors are linearly independent. The span of 3 vectors has a dimension of at most 3. For example, $(0, 0, 0, 1)$ is not in the span of these vectors since

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 4 & 0 \\ 3 & 1 & 4 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

is inconsistent.

8. Let $c_1(\mathbf{v} - \mathbf{w}) + c_2(2\mathbf{v} + \mathbf{w}) = \mathbf{0}$. (We need to show that $c_1 = c_2 = 0$ which implies that the vectors are linearly independent).

$$\begin{aligned} c_1(\mathbf{v} - \mathbf{w}) + c_2(2\mathbf{v} + \mathbf{w}) &= \mathbf{0} \\ \Rightarrow \mathbf{v}(c_1 + 2c_2) + \mathbf{w}(c_2 - c_1) &= \mathbf{0} \end{aligned}$$

Since $\{\mathbf{v}, \mathbf{w}\}$ is linearly independent, $c_1 + 2c_2 = 0$ and $c_2 - c_1 = 0$. This gives the homogeneous system of equations,

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

which implies that $c_1 = c_2 = 0$ as needed,